Unbox, Sort, and Stow: Closed-Loop Mobile Manipulation for Grocery Organization

Problem Statement

In aging households, daily food management tasks such as unpacking groceries and organizing items into storage areas can become physically demanding and require sustained attention and planning. These tasks involve coordinated mobility, object manipulation, perception, and task-level reasoning.

This project aims to develop a mobile manipulation system that autonomously navigates to a grocery pickup location, extracts items from an open container, and places them into designated storage bins (representing refrigerator). Storage decisions are made through a semantic inference module that maps detected object classes to appropriate storage policies (e.g., refrigeration, freezing, room temperature) using a structured knowledge base, which are then translated into spatial target zones.

The goal is to build an integrated ROS system that combines navigation, perception, manipulation, semantic reasoning, and closed-loop verification within a unified task execution pipeline.

Demo Task Description

The robot will perform the long-chain task in the following sequence:

1. Navigate to a predefined **Pickup Zone** where an open grocery container is located.
2. Perceive and detect objects inside the container using RGB-D sensing.
3. Select a target object based on reachability and confidence.
4. Plan and execute a grasp using MoveIt!.
5. Navigate to a predefined **Stow Zone** (simulated refrigerator).
6. Perceive the designated storage bin.
7. Plan and execute a placement motion.
8. Verify placement success and update the internal inventory state.
9. Repeat until all objects have been processed.

For safety and reproducibility, the grocery container will be pre-opened. The focus of the task is object extraction and storage management rather than deformable package manipulation.

System Overview

The system will be deployed on a mobile manipulation platform and fully controlled through ROS2. The architecture is composed of the following major components:

1. **Mobile Base (Nav2 stack)**
   The mobile base will use the ROS2 Nav2 stack for 2D localization, mapping, and navigation between the Pickup and Stow zones. The base will support waypoint navigation and docking behaviors.
2. **Manipulation (MoveIt!)**
   The robotic arm will be controlled via MoveIt2 for motion planning, collision-aware trajectory generation, and execution of grasp and placement actions. A planning scene will model the pickup container, table surfaces, and storage bins.
3. **Perception Module (RGB-D camera)**
   An RGB-D camera mounted on the platform will detect objects and estimate their pose relative to the robot frame. AprilTags may be used initially to ensure robustness, with the possibility of extending to markerless detection.
4. **High-Level Task Supervisor**
   A custom task-level controller (state machine or behavior tree) will coordinate navigation, perception, manipulation, semantic reasoning, and failure recovery.
5. **Semantic Inference and Inventory Manager (Custom Nodes)**
   A semantic inference module will map detected object classes to storage policies using a structured knowledge base. An inventory manager node will maintain internal state tracking detected, picked, and stowed objects, enabling consistent task execution and recovery strategies.

Required Capabilities

To accomplish the demo task, the robot needs to have:

- Accurate 2D navigation and docking near task zones
- RGB-D perception and object pose estimation
- Grasp planning with collision checking
- Closed-loop verification of grasp and placement
- Task-level decision-making and sequencing
- Failure recovery and re-perception strategies

Nodes

The following nodes will be implemented from scratch:

- **Object Selector Node:** Chooses the next object to grasp based on reachability and confidence.

- **Inventory Manager Node:** Tracks object states (detected, picked, stowed) and semantic storage rules.
- **Grasp Refinement Node:** Performs pose refinement and generates grasp candidates.
- **Task Supervisor Node:** Coordinates navigation, manipulation, and perception through a custom state machine or behavior tree.

Evaluation

System performance will be evaluated using the following metrics:

- **Task Success Rate**
  Percentage of objects correctly placed into their designated storage zones according to the semantic storage policy.
- **Grasp Success Rate**
  Ratio of successful grasps to total grasp attempts, including retries.
- **Failure Recovery Rate**
  Percentage of failed navigation or manipulation actions that are successfully resolved through re-perception, re-planning, or retry strategies.

These metrics evaluate both subsystem reliability and the robustness of the integrated closed-loop pipeline.